Aga Khan University Examination Board

Notes from E-Marking Centre on HSSC-II Business Statistics Examination April/ May 2019

Introduction

This document has been produced for the teachers and candidates of Higher Secondary School Certificate (HSSC-II) Business Statistics. It contains comments on candidates' responses to the 2019 HSSC-II Examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that require candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

Key observations

This year candidates did not perform well in questions based on price index, permutation and combination were not well attempted. Weaker responses revealed that candidates had problems with finding median and mode for grouped data and probability.

Detailed Comments:

Constructed Response Questions (CRQs)

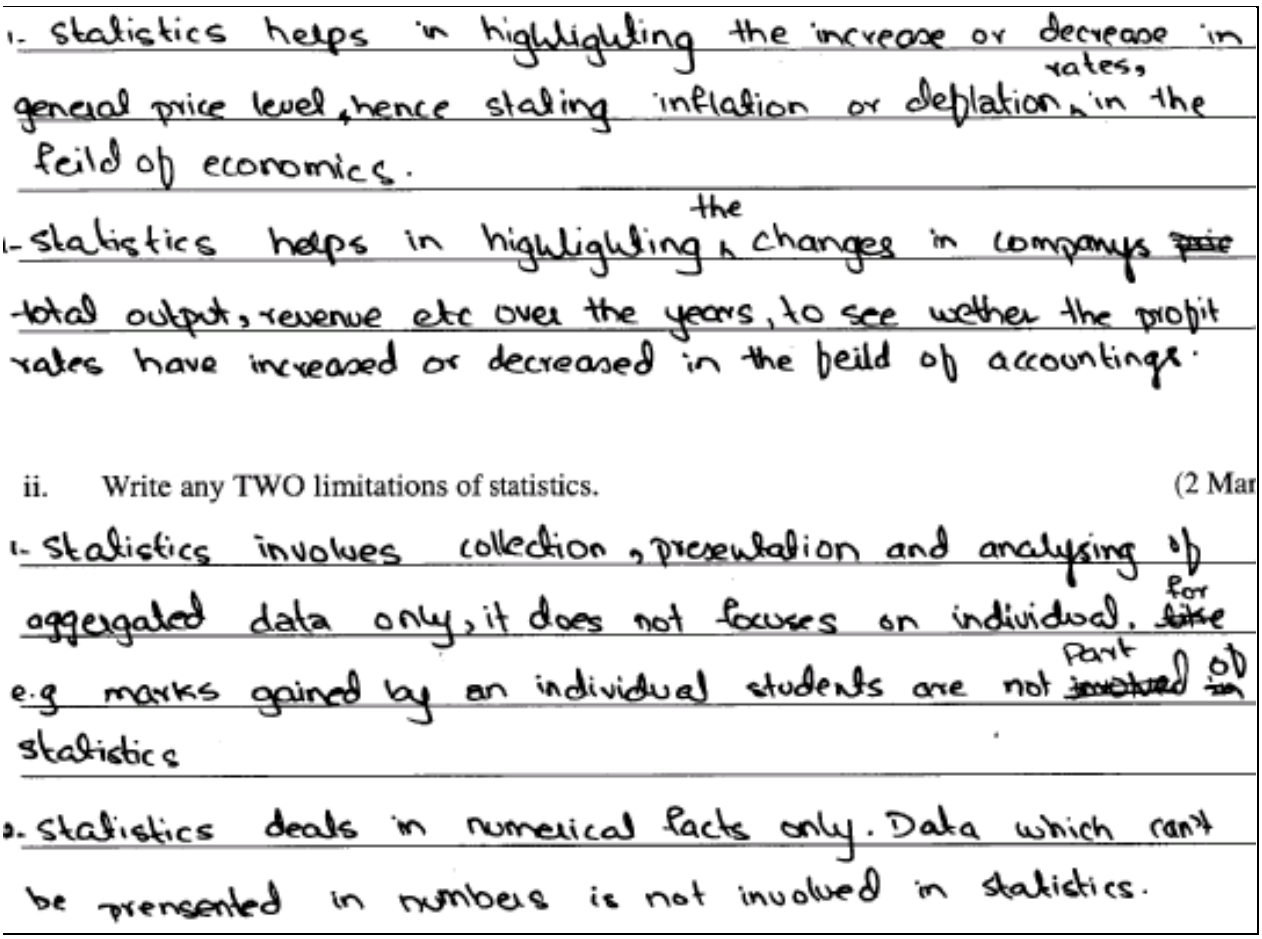
Question 1:

i. Write any TWO points to highlight the importance of statistics in different fields.

ii. Write any TWO limitations of statistics.

Better responses indicated that in part **i**, the candidates were able to highlight the importance of statistics in different fields. In part **ii**, candidates enlisted correct limitations of statistics, as given in the following example.

Example:



1- statistics helps in highlighting the increase or decrease in general price level, hence stating inflation or deflation rates, in the feild of economics.

1- statistics helps in highlighting the changes in companys ~~price~~ total output, revenue etc over the years, to see wether the profit rates have increased or decreased in the feild of accountings.

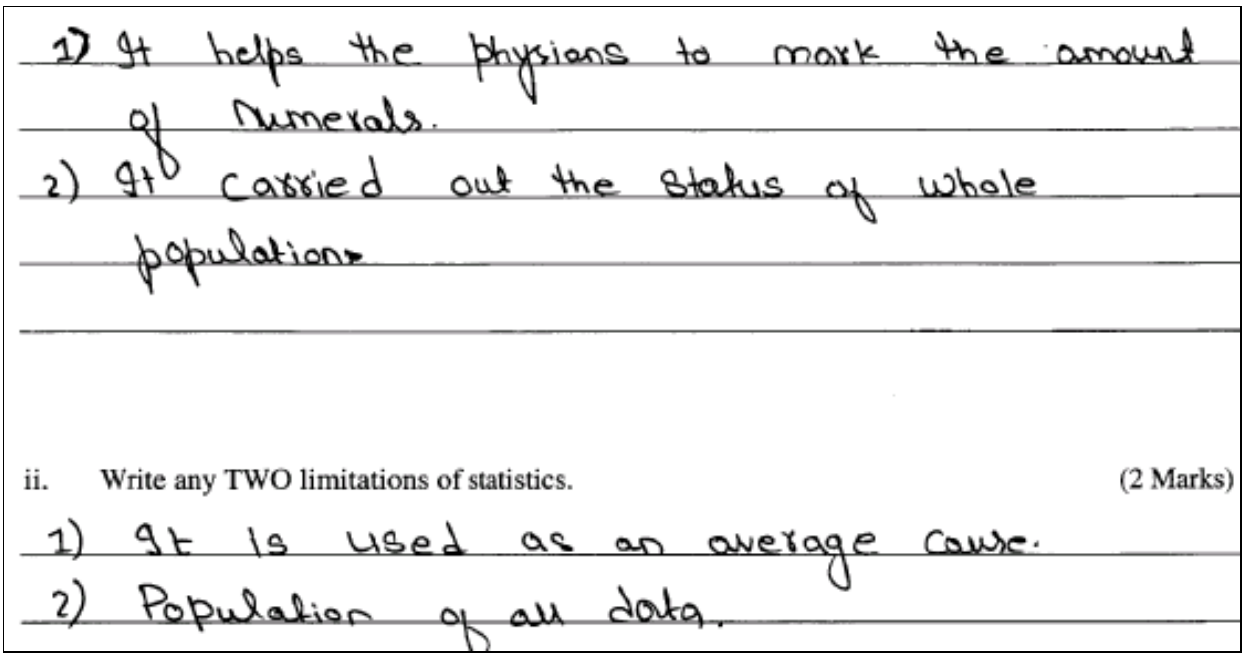
ii. Write any TWO limitations of statistics. (2 Mar

1- statistics involves collection, presentation and analysing of aggregated data only, it does not focuses on individual. for ~~like~~ e.g marks gained by an individual students are not Part ~~involved~~ of in statistics

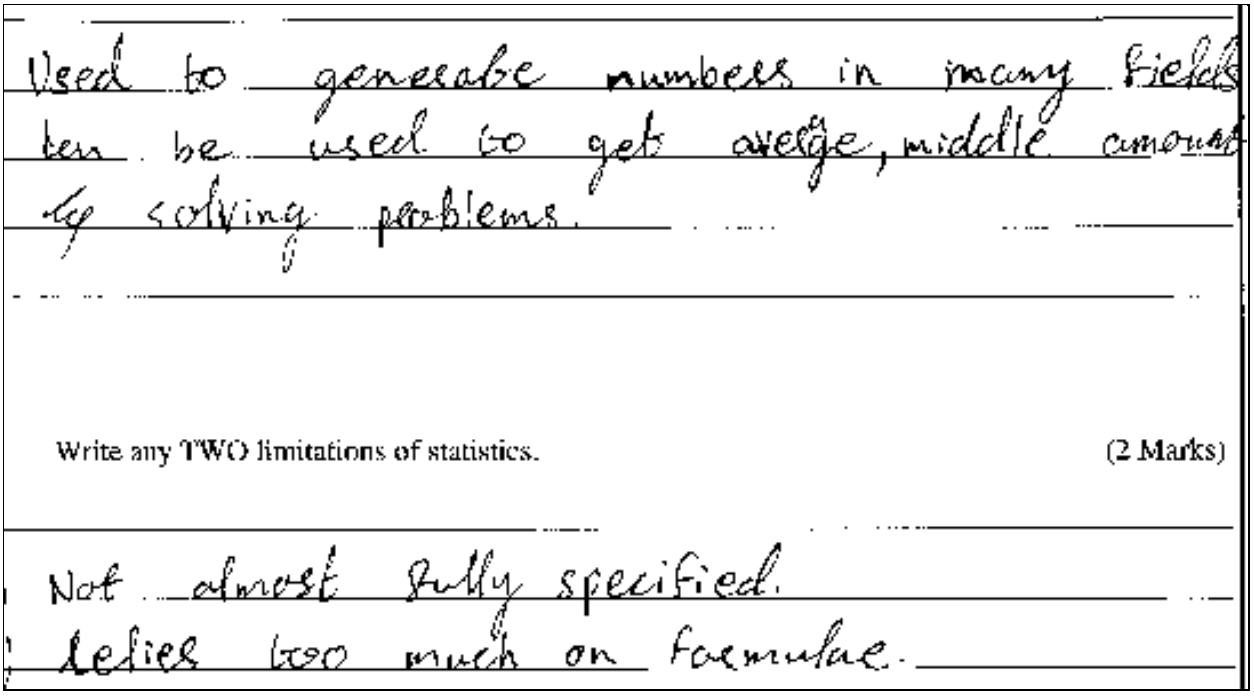
2. statistics deals in numerical facts only. Data which can't be presented in numbers is not involved in statistics.

Weaker responses revealed that in part i, candidates lacked knowledge about the importance of statistics in different fields data and failed to highlight its importance. In Part ii the responses of the candidates reflected that candidates had no idea about the limitations of statistics and wrote random responses as cited in the given example.

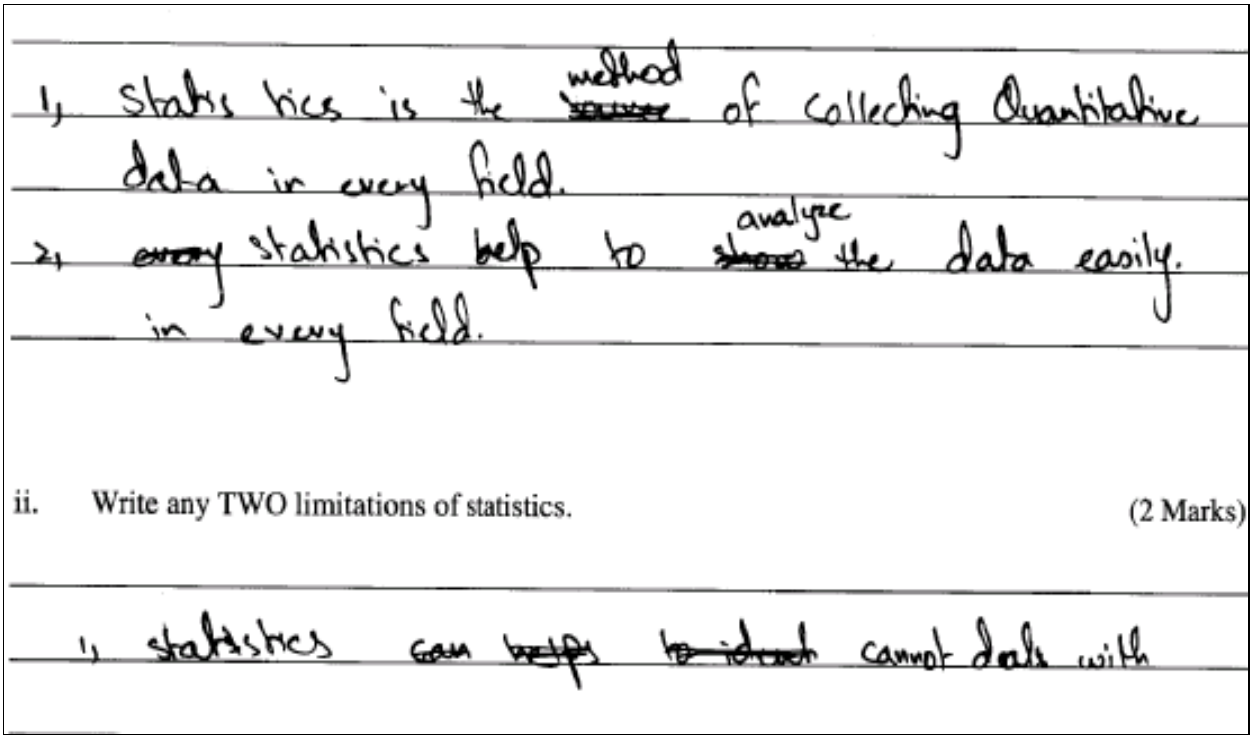
Example 1:



Example 2:



Example 3:



1, Statistics is the ~~source~~ method of collecting Quantitative data in every field.

2, ~~every~~ statistics help to ~~show~~ analyse the data easily. in every field.

ii. Write any TWO limitations of statistics. (2 Marks)

1, statistics can ~~help~~ ~~to~~ ~~show~~ cannot deals with

Question 2:

In an under 19 cricket camp, the weights of 25 players were recorded as follows:

35, 40, 40, 40, 45, 45, 45, 44, 37, 37, 36, 50, 50, 38, 38, 39, 41, 46, 47, 47, 48, 43, 43, 42, 42

i. Use the given table to construct the frequency distribution of the given data.s)

ii. Also complete the column for cumulative frequency and class mark.)

Weight (kg)	Number of Participants	Cumulative Frequency	Class Mark
35 – 37			
38 – 40			
41 – 43			
44 – 46			
47 – 49			
50 – 52			

Better responses showed that the candidates were well aware of the concepts of frequency distribution. They found the frequency of the each class interval correctly. They also find the class mark and required cumulative frequency aptly.

Example:

35, 40, 40, 40, 45, 45, 45, 44, 37, 37, 36, 50, 50, 38, 38, 39, 41, 46, 47, 47, 48, 43, 43, 42, 42

i. Use the given table to construct the frequency distribution of the given data. (2 Marks)

50 - 52

ii. Also complete the column for cumulative frequency and class mark. (2 Marks)

Weight (kg)	Number of Participants	Cumulative Frequency	Class Mark
35 – 37	4	4	$\frac{35+37}{2} = 36$
38 – 40	6	10	$\frac{38+40}{2} = 39$
41 – 43	5	15	$\frac{41+43}{2} = 42$
44 – 46	5	20	$\frac{44+46}{2} = 45$
47 – 49	3	23	$\frac{47+49}{2} = 48$
50 – 52	2	25	$\frac{50+52}{2} = 51$

As the data is in ascending order, cumulative frequency less than is constructed. Class marks is the midpoint of the class interval, calculated by taking the average of the class intervals.

CF = 25

Weaker responses displayed that candidates lacked in the concepts being tested in the question and failed to find the required frequency distribution, cumulative frequency and class mark. They used wrong formulae to find the required quantities, although formulae were fairly simple.

Example:

Weight (kg)	Number of Participants	Cumulative Frequency	Class Mark
35 – 37	5	34·5 – 37·5	36
38 – 40	5	37·5 – 40·5	39
41 – 43	3	40·5 – 43·5	42
44 – 46	2	43·5 – 46·5	45
47 – 49	5	46·5 – 49·5	48
50 – 52	5	49·5 – 52·5.	51

Example 2:

Weight (kg)	Number of Participants	Cumulative Frequency	Class Mark
35 – 37	4	0·16	34·5 – 37·5
38 – 40	6	0·24	37·5 – 40·5
41 – 43	5	0·2	40·5 – 43·5
44 – 46	5	0·2	43·5 – 46·5
47 – 49	3	0·12	46·5 – 49·5
50 – 52	2	0.08	49·5 – 52·5
total	25		

Question 3:

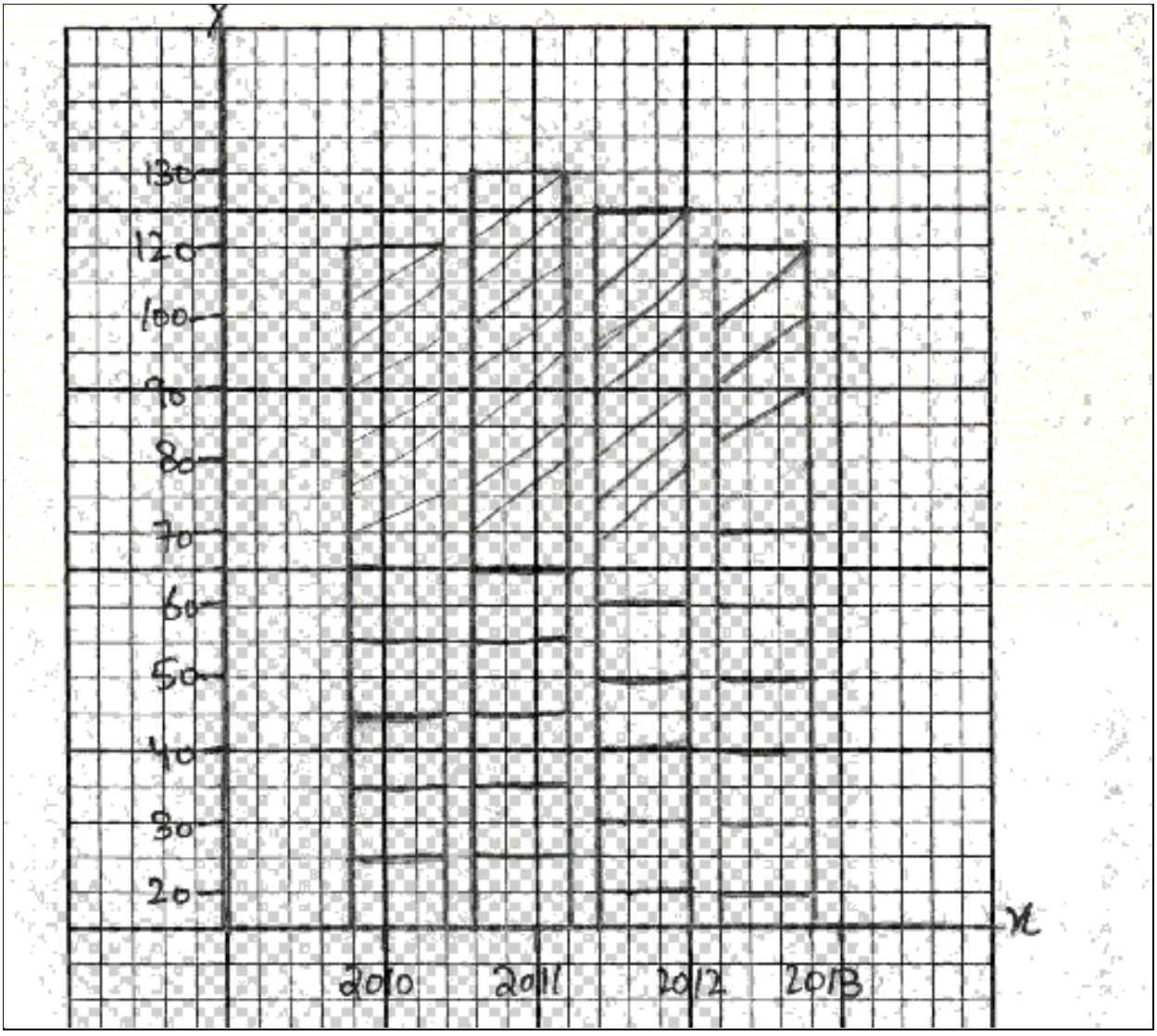
The given table shows the yields, in thousand kilograms, of wheat, rice and gram. These commodities are produced at a certain agricultural farm from years 2010 to 2013.

Year	Wheat (kg)	Rice (kg)	Gram (kg)
2010	65	35	20
2011	65	40	25
2012	60	40	25
2013	70	30	20

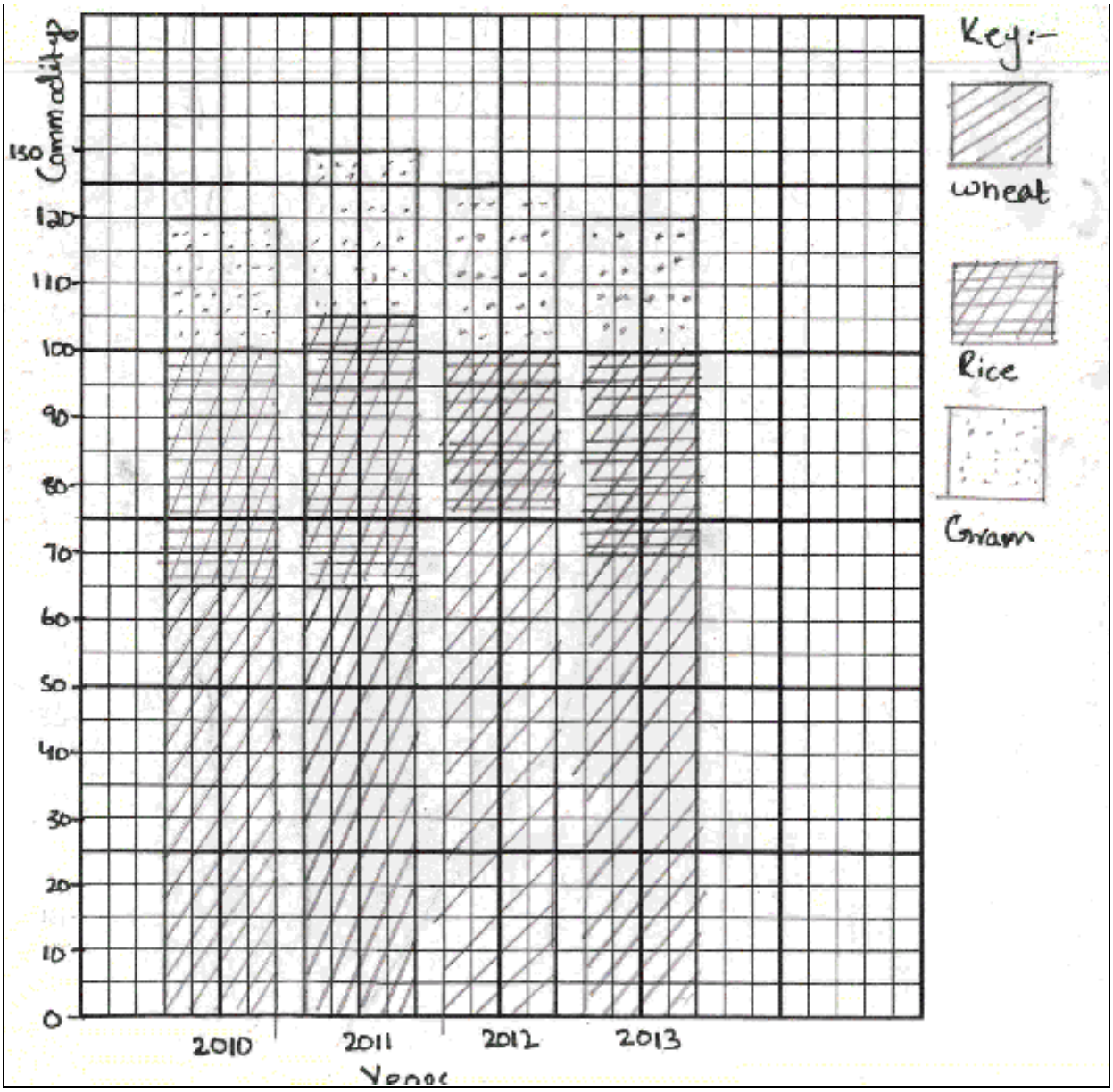
On the given graph, construct a component/ subdivided bar chart to represent the data.

Better responses reflected that candidates were skilfully able to construct the component bar chart for the given data. The candidates took appropriate scale on both the axes and clearly labelled the axes to represent the data through component/ subdivided bar chart effectually.

Example 1:

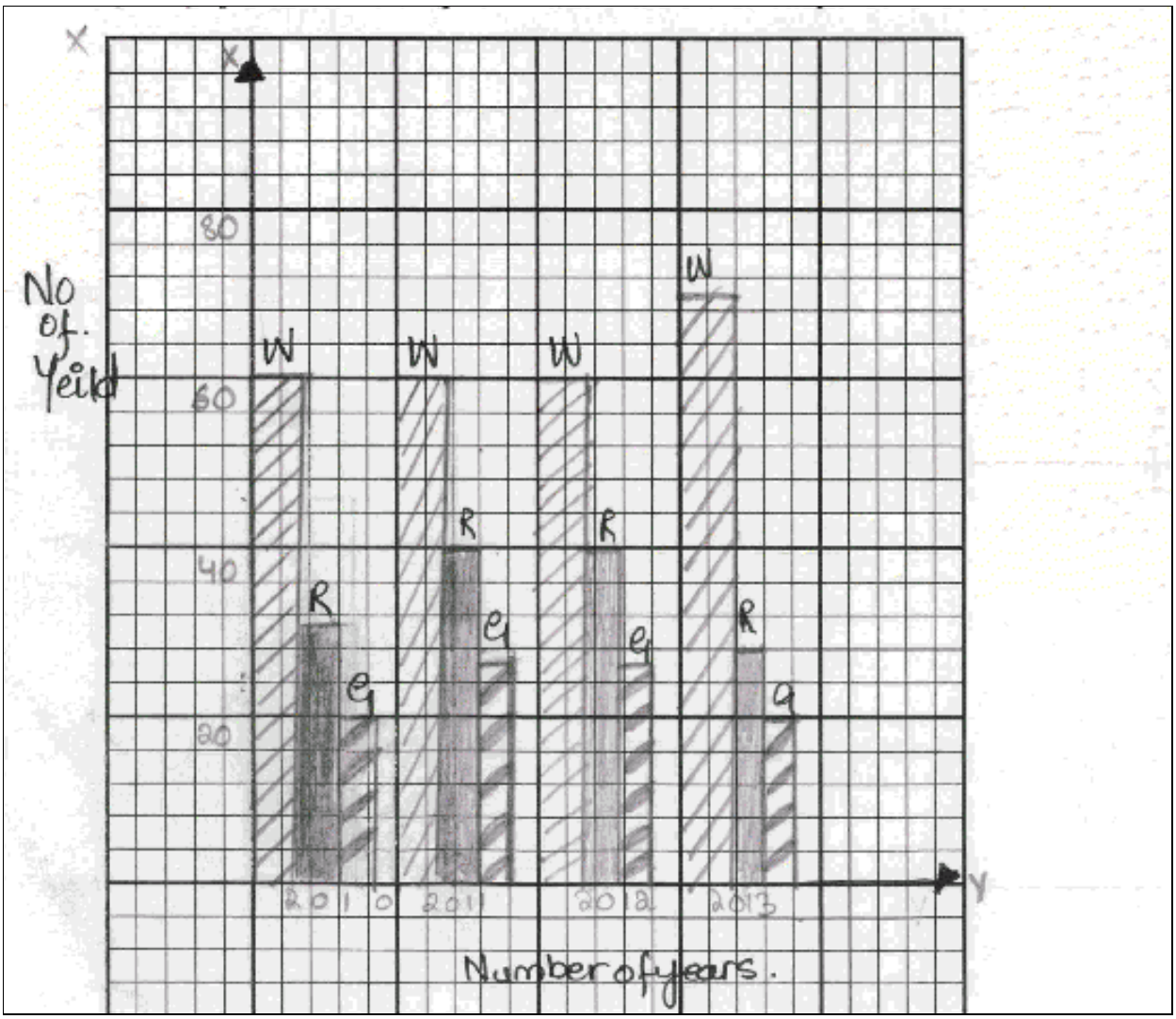


Example 2:

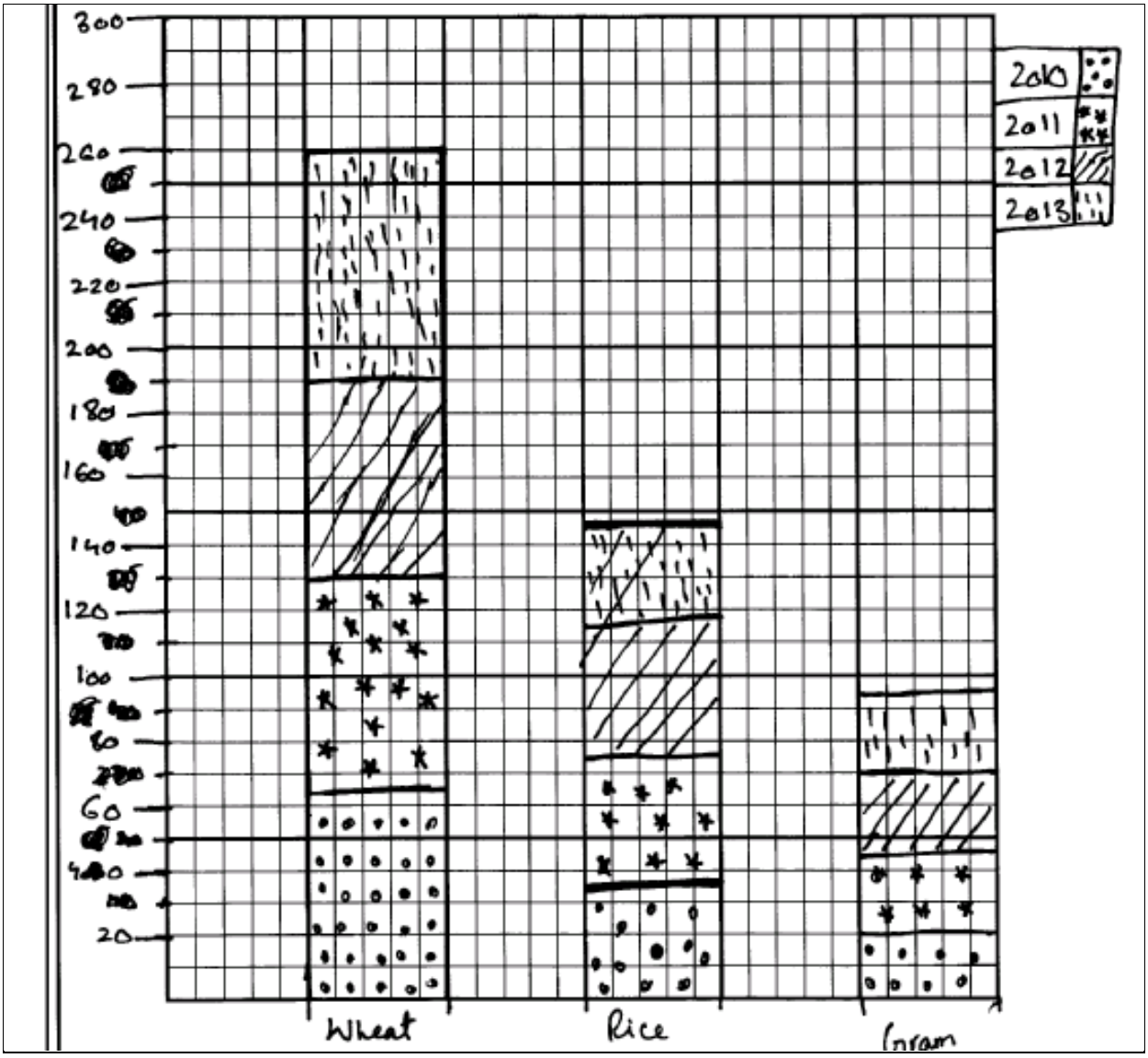

Commodity
150
120
110
100
90
80
70
60
50
40
30
20
10
0
2010
2011
2012
2013
Years
Key:-
wheat
Rice
Gram

Weaker responses indicated that candidates were confused between multiple bars diagram and subdivided bar Diagram and mostly they constructed multiple bars diagram. They failed to label the axes and unable to choose the proper scale to construct the required diagram for precise representation of data.

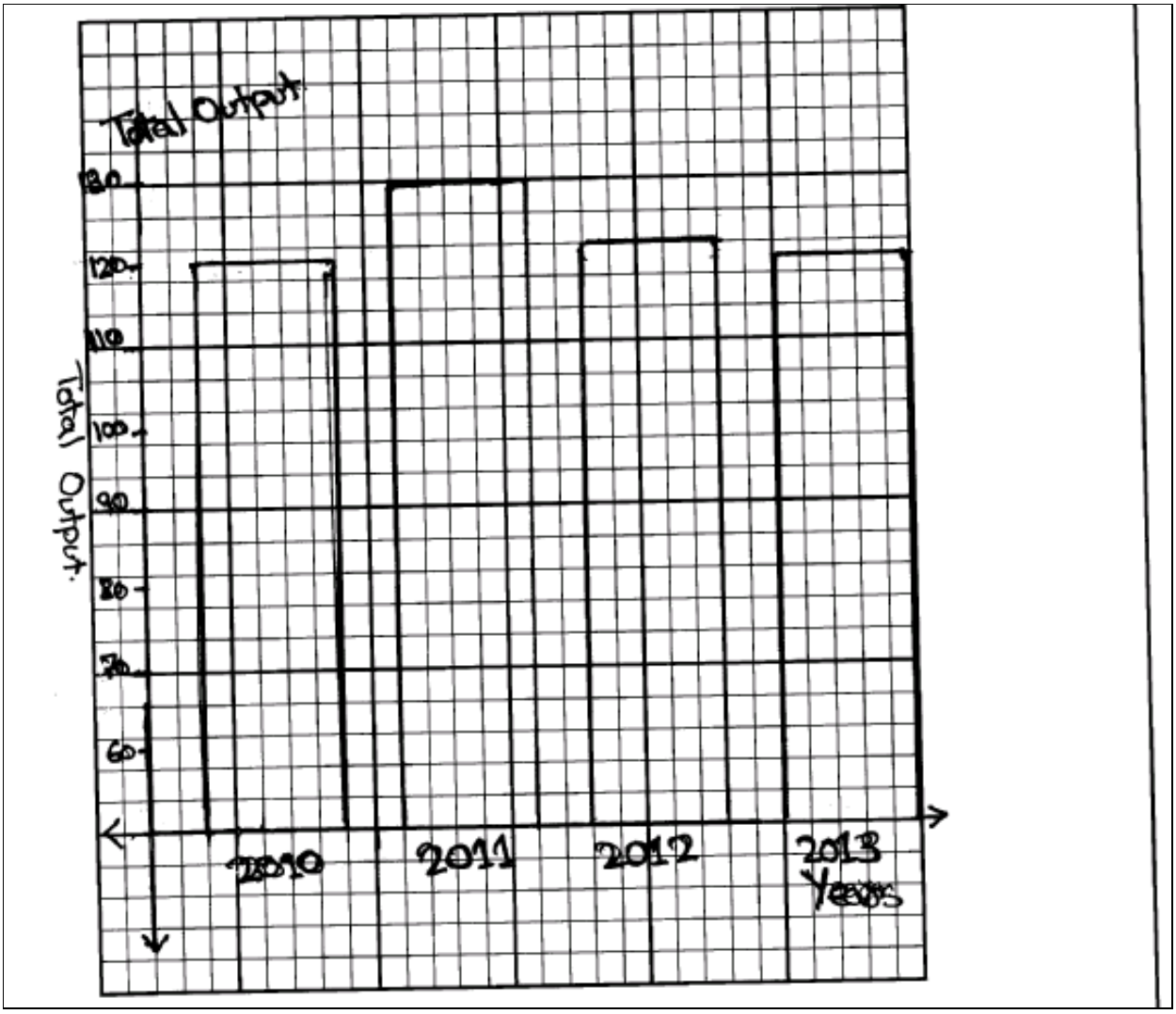
Example 1:



Example 2:


300
280
260
240
220
200
180
160
140
120
100
80
60
40
20
2010
2011
2012
2013
Wheat
Rice
Gram

Example 3:


Total Output
130
120
110
100
90
80
70
60
Total Output
2010
2011
2012
2013
Years

Question 4a:

In the given table, Bilal recorded the time in seconds of 30 people to complete a race.

Time (s)	**Number of People**	
51 – 55	4	
56 – 60	10	
61 – 65	11	
66 – 70	5	

Label and fill the empty column to find the median of this data.

This was generally a well attempted question.

Better responses reflected that candidates found cumulative frequency of the frequency distribution and correctly identified the median class. They wrote the formula and identified its elements from the given data. Subsequently they calculated median correctly.

Example:

Time (s)	Number of People	C.f
51 – 55	4	4
56 – 60	10	14
61 – 65	11	25
66 – 70	5	30

Label and fill the empty column to find the median of this data.

$\frac{\Sigma f}{2} = \frac{30}{2} = 15$

$\bar{x} = L + \frac{h}{f}\left(\frac{\Sigma f}{2} - C.f\right)$

$\bar{x} = 60.5 + \frac{5}{11}(15 - 14)$

$\bar{x} = 60.5 + 0.4545$

$\bar{x} = 60.9545$

Median = 60.9545

Data:

$L = 60.5$

$h = 5$

$f = 11$

$C.f = 14$

$\bar{x} = ?$

Weaker responses revealed that the candidates made different types of mistakes which are reflective of their misconception or lack of understanding of median of grouped data. Few of the mistakes noted are as follows:

- They were unable to find the cumulative frequency correctly
- They made mistakes in identification of the median class interval
- They failed to apply the formula correctly
- They made mistakes in calculations
- Three examples are given to cite these types of mistakes.

Example 1:

Time (s)	Number of People	cf
51 – 55	4	0·1333
56 – 60	10	0·333
61 – 65	11	0·366
66 – 70	5	0·166

Label and fill the empty column to find the median of this data.

$$L + \frac{h}{F}\left(\frac{\Sigma f}{2} - cf\right)$$

$$61 + \frac{5}{36}\left(\frac{30}{2} - 0{\cdot}9998\right)$$

$$= 167{\cdot}4665$$

Example 2:

Time (s)	Number of People	cf
51 – 55	4	0.1333
56 – 60	10	0.333
61 – 65	11	0.366
66 – 70	5	0.166

Label and fill the empty column to find the median of this data.

$$L + \frac{h}{f}\left(\frac{\Sigma f}{2} - cf\right)$$

$$61 + \frac{5}{36}\left(\frac{30}{2} - 0.9998\right)$$

$$= 167.4665$$

Example 3:

Time (s)	Number of People	Class Boundaries
51 – 55	4	50.5 — 55.5
56 – 60	10	55.5 — 60.5
61 – 65	11	60.5 — 65.5
66 – 70	5	65.5 — 70.5

Label and fill the empty column to find the median of this data

$$\tilde{x} = l + \frac{h}{f}\left(\frac{n}{2} - c\right)$$

$$\tilde{x} = 55.5 + \frac{5}{10}\left(\frac{30}{2} - 14\right)$$

$$\tilde{x} = \frac{555 + 50}{10}\left(\frac{30}{2} - 14\right)$$

$$\tilde{x} = \frac{605}{10}\left(\frac{30}{2} - 14\right)$$

$$\tilde{x} = \frac{605}{10}\left(\frac{60 - 28}{2}\right)$$

$$\tilde{x} = \frac{605}{10} \times \frac{32}{2}$$

$$\tilde{x} = \frac{1210}{320}$$

Question 4b:

The given data shows the ages of 30 people working in a department. Label and fill empty column of the table to calculate the mode of the data and interpret your answer.

Age (year)	**Number of People**	
31 – 40	9	
41 – 50	5	
51 – 60	10	
61 – 70	6	

Better responses reported that the candidates were able to find the mode of the given ungrouped data. They choose the modal class based on the highest frequency in the given data. They applied the formula correctly and made right calculation.

Example 1:

Age (year)	Number of People	C·B
31 – 40	9	30·5 – 40·5
41 – 50	5	40·5 - 50·5
51 – 60	10	50·5 - 60·5
61 – 70	6	60·5 - 70·5

$\hat{x} = L + \frac{f_m - f_1}{2f_m - f_1 - f_2} \times h$

$\hat{x} = 50.5 + \frac{10 - 5}{2(10) - 5 - 6} \times 10$

$\hat{x} = 50.5 + \frac{5}{9} \times 10$

$\hat{x} = 50.5 + 5.56$

$\hat{x} = 56.06$

Mode = 56.06

Data:

$L = 50.5$

$f_m = 10$

$f_1 = 5$

$f_2 = 6$

$h = 10$

The mode of given data which means the one with highest frequency belongs to 3 class, and the calculated mode is 56.06, for the people in a department.

Weaker responses showed that the candidates were unable to find the mode correctly and showed confusions in finding the mode. Similarly they were failed to decide that about the modal class of the given data or wrote wrong formula. The weaker responses also indicated that candidates made calculation mistakes.

Example 1:

Age (year)	Number of People	class boundaries
31 – 40	9	30.5 – 40.5
41 – 50	5	40.5 – 50.5
51 – 60	10	50.5 – 60.5
61 – 70	6	60.5 – 70.5

Mode, most frequent or repeated value.

51 – 60 = 10.

This shows the class interval 51 – 60 have 10 people. 10 people age between 51 to 60 which makes 33.33.1. of the total.

Example 2:

Age (year)	Number of People	x	C.F	
31 – 40	9	35.5	9	
41 – 50	5	45.5	14	
51 – 60	10	55.5	24	
61 – 70	6	65.5	30	

Highest frequency

$$\hat{x} = L + \frac{f_m - f_1}{2f_m - f_1 - f_2} \times h$$

$$= 50.5 + \frac{10 - 5}{2(10) - 5 - 6} \times 10$$

$$= 53.833$$

Example 3:

Age (year)	Number of People	c.b
31 – 40	9	30.5 – 40.5
41 – 50	5	40.5 – 50.5
51 – 60	10	50.5 – 60.5
61 – 70	6	60.5 – 70.5

mode: $L + \left(\frac{f_m - f_1}{2f_m - f_1 - f_2}\right) \times h$

$50.5 + \left(\frac{10 - 5}{2(10) - (5) - (6)}\right) \times 10$

$50.5 \left(\frac{5}{20 - (30)}\right) \times 10$

$50.5 + \left(\frac{5}{600}\right) \times 10$

$50.5 \quad 0.0083 \times 10$

$= 0.83$

50.50×10

505.083

Question 5:

In a nursery, the lengths of 80 plants of certain species were measured in centimetres. These measurements were recorded in the given table.

Length of Plant (cm)	**Number of Plants**	**Class Mark**		
15 – 19	12	17		
20 – 24	17	22		
25 – 29	28	27		
30 – 34	13	32		
35 – 39	10	37		
Total	**80**	**-**	**-**	

For the given data, the mean height is 26.5 cm. Fill the empty column of the table to calculate mean deviation of the heights of the plants.

Better responses indicated that the candidates were able to understand the scenario and apply their skills in finding the mean deviation as per the given data. They correctly labelled the unknown columns as $\left|x_i - \bar{x}\right|$ and $f_i \times \left|x_i - \bar{x}\right|$, found the values $\sum \left|x_i - \bar{x}\right|$, and $\sum f \times \left|x_i - \bar{x}\right|$ and substituted these values in the formula $MD = \dfrac{\sum f_i . \left|x_i - \bar{x}\right|}{\sum f_i}$ to find the required mean deviation of the data.

Example:

Length of Plant (cm)	Number of Plants	Class Mark	$\lvert x - \bar{x} \rvert$	$f\lvert x - \bar{x} \rvert$
15 – 19	12	17	$\lvert 17 - 26.5 \rvert = 9.5$	$12 \times 9.5 = 114$
20 – 24	17	22	$\lvert 22 - 26.5 \rvert = 4.5$	$17 \times 4.5 = 76.5$
25 – 29	28	27	$\lvert 27 - 26.5 \rvert = 0.5$	$28 \times 0.5 = 14$
30 – 34	13	32	$\lvert 32 - 26.5 \rvert = 5.5$	$13 \times 5.5 = 71.5$
35 – 39	10	37	$\lvert 37 - 26.5 \rvert = 10.5$	$10 \times 10.5 = 105$
Total	**80**	-	-	

For the given data, the mean height is 26.5 cm. Fill the empty column of the table to calculate mean deviation of the heights of the plants.

Mean Deviation $= \dfrac{\sum f \lvert x - \bar{x} \rvert}{\sum f}$

$= \dfrac{381}{80}$

$= 4.7625$ is the mean deviation

Weaker responses indicated that the candidates did not have the concept of men deviation for groped data. Although the mean of the distribution was given in the question but weaker responses reported that candidates calculated the mean again. Moreover, they were unable to substitute values in the formula accurately and as a result they were failed to find the value of mean deviation.

Example 1:

Length of Plant (cm)	Number of Plants f	Class Mark x	fx	
15 – 19	12	17	204	
20 – 24	17	22	374	
25 – 29	28	27	756	
30 – 34	13	32	416	
35 – 39	10	37	370	
Total	**80**	-	-	

For the given data, the mean height is 26.5 cm. Fill the empty column of the table to calculate mean deviation of the heights of the plants.

formula = $\frac{\Sigma |fx - \bar{x}|}{\Sigma f}$

$= \frac{2120 - 26.5}{80}$

$\bar{x} = \frac{fx}{f} = \frac{2120}{80}$

$= 26.5$.

4 $\bar{x} = 26.16$

Example 2:

Length of Plant (in cm)	Number of Plants	Class Mark x	$x - \bar{x}$	$\lvert x - \bar{x} \rvert$
15 – 19	12	17	$12 - \frac{16}{80} = -04$	04
20 – 24	17	22	$17 - 16 = 01$	01
25 – 29	28	27	$28 - 16 = 12$	12
30 – 34	13	32	$13 - 16 = -03$	03
35 – 39	10	37	$10 - 16 = -6$	06
Total	80	-	-	$\Sigma\lvert x - \bar{x} \rvert = 120$

For the given data, the mean height is 26.5 cm. Fill the empty column of the table to calculate mean deviation of the heights of the plants.

$$\text{Mean deviation} = \frac{\Sigma f \lvert x - \bar{x} \rvert}{\Sigma f}$$

$$= \frac{80(120)}{80}$$

$$= \frac{9600}{80}$$

$$= 120$$

$$\bar{X} = \frac{\Sigma f_i}{n} = \frac{80}{5} = 16$$

Example 3:

Length of Plant (cm)	Number of Plants	Class Mark	$(x-\bar{x})$	$\lvert x-\bar{x}\rvert$
15 – 19	12	17	$(17-26.5)=-9.5$	$\lvert -9.5\rvert = 9.5$
20 – 24	17	22	$(22-26.5)=-4.5$	$\lvert -4.5\rvert = 4.5$
25 – 29	28	27	$(27-26.5)=0.5$	$\lvert 0.5\rvert = 0.5$
30 – 34	13	32	$(32-26.5)=5.5$	$\lvert 5.5\rvert = 5.5$
35 – 39	10	37	$(37-26.5)=10.5$	$\lvert 10.5\rvert = 10.5$
Total	**80**	-	-	$\Sigma\lvert x-\bar{x}\rvert = 30.5$

For the given data, the mean height is 26.5 cm. Fill the empty column of the table to calculate mean deviation of the heights of the plants.

$$M.D = \frac{\Sigma|x-\bar{x}|}{n}$$

$$= \frac{30.5}{5}$$

$$= 6.1$$

Question 6ai:

Write any TWO steps of construction of price index.

Question 6aii:

Describe any TWO steps of construction of price index.

Better responses accurately wrote the two steps of construction of price index. In the second part they wrote the details of these two steps correctly. They were clear about the demand of the question.

Example:

Two steps of construction of price index;

① Purpose

② Base method

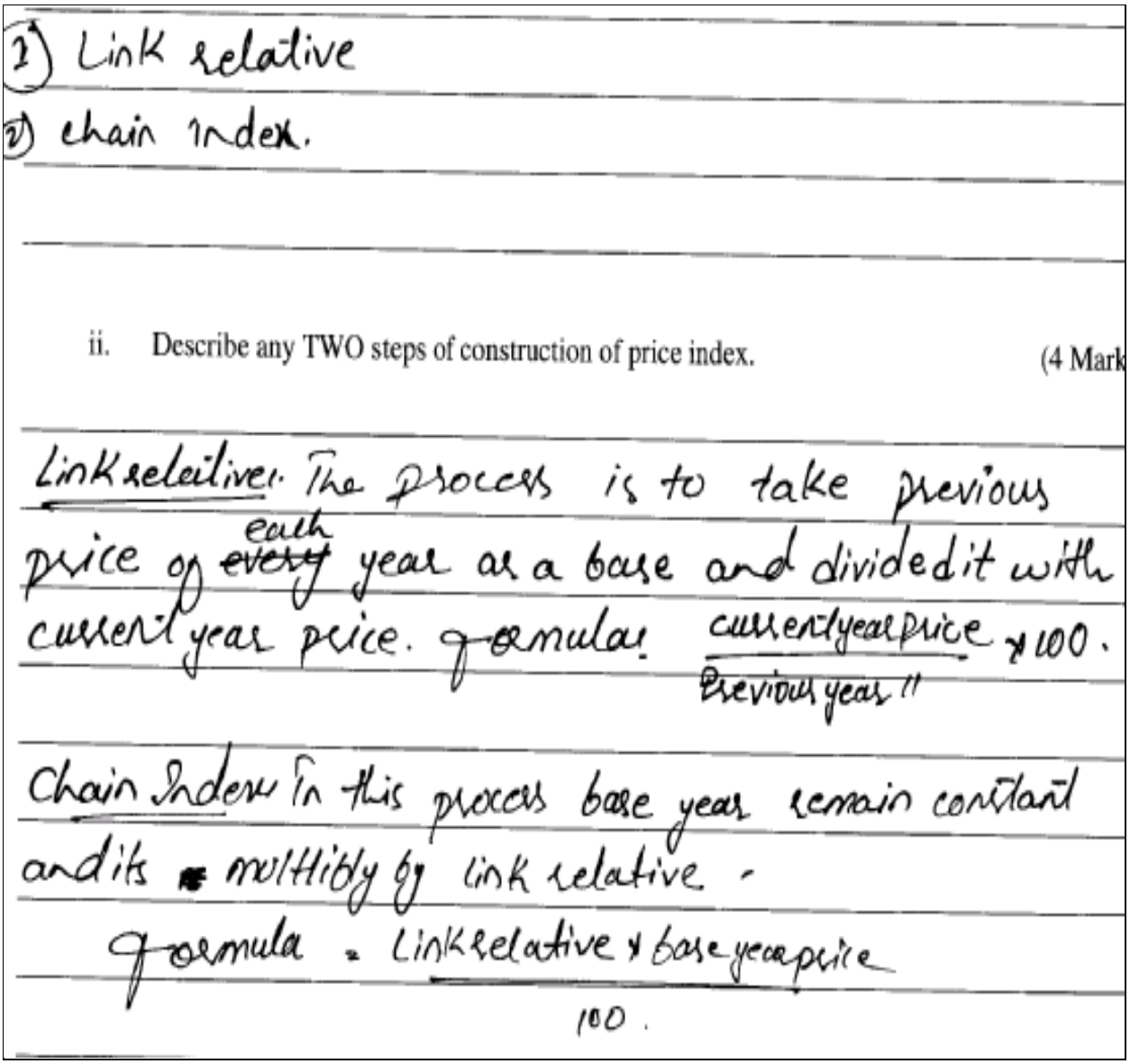
ii. Describe any TWO steps of construction of price index. (4 M

↳ Purpose is an important step in construction of price index. The purpose should be determined. ~~for which~~ What changes are to be measured in a variable or group of variables.

↳ Base method is of two type ① fixed based in which average price of a particular year is taken as base year. ② chain base method in which base year changes every year.

Weaker responses displayed that candidates did not know the steps of construction of index number and wrote unnecessary and irrelevant responses as evident from the following examples.

Example 1:



1) Link relative
2) chain index.

ii. Describe any TWO steps of construction of price index. (4 Mark

Link relative: The process is to take previous price of ~~every~~ each year as a base and divided it with current year price. formula: $\frac{\text{current year price}}{\text{Previous year "}} \times 100$.

Chain Index: In this process base year remain constant and its ~~=~~ multiply by link relative.

formula = $\frac{\text{Link relative} \times \text{base year price}}{100}$.

Example 2:

1) column according to method

2) solve with help of formula

ii. Describe any TWO steps of construction of price index. (4 Marks

1) first you have to Draw a column of year, price and a column base year. If the method is price relative then then a base year is also given which is called fixed base and if the method is link relative then no base year column is required but you have pro Draw a column of link relative. If the question has also a requirement of chain indices then column of chain indices must be made.

2) After making column, we solve price relative by dividing current year from ~~fixed base year~~ fixed base year and multiply by hundred and so on. In the case of link relative we divide current year from preceeding year and multiply by 100

Question 6b:

For the given data in the table, calculate Paasche's index number using base year 2010. Also interpret your result.

	2010		2015			
Commodity	**Quantity (kg)**	**Price/ kg (Rs)**	**Quantity (kg)**	**Price/ kg (Rs)**		
Wheat	100	10	120	13		
Rice	50	100	40	110		
Cotton	35	148	40	160		
Butter	12	600	10	750		
Mutton	48	500	35	600		
Total	-	-	-	-		

Better responses indicated that the candidates had correct knowledge of Paasche's index number and wrote the formula correctly. From the given table, they found the values of $p_n q_n$, $p_o q_n$ and found the sum correctly. The values of $\sum p_n q_n$ and $\sum p_o q_n$ substituted in the formula to find the correct answer.

Example:

Commodity	2010		2015		$P_n q_n$	$P_o q_n$
	Quantity (kg) q_o	Price/ kg (Rs) P_o	Quantity (kg) q_n	Price/ kg (Rs) P_n		
Wheat	100	10	120	13	1560	1200
Rice	50	100	40	110	4400	4000
Cotton	35	148	40	160	6400	5920
Butter	12	600	10	750	7500	6000
Mutton	48	500	35	600	21000	17500
Total	-	-	-	-	40860	34620

Paasche index number $= \frac{\Sigma P_n q_n}{\Sigma P_o q_n} \times 100$ ∵ Paasche formula deals with current year weight index number.

$\Sigma P_n q_n = 40860$, $\Sigma P_o q_n = 34620$

$P_{on} = \frac{40860}{34620} \times 100$

$P_{on} = 118.02\%$

In 2015 the price and quantity of given commodities are increased by 18.02% from 2010 by taking weigh of current year.

Weaker responses displayed that most of the candidates were able to identify and label the four columns as p_o, q_o, p_n and q_n, but they could not figure out what was required in the last two columns. Use of incorrect formula was a frequent error too. Few more errors are evident from the given examples.

Example 1:

Commodity	2010 Quantity (kg)	2010 Price/kg (Rs)	2015 Quantity (kg)	2015 Price/kg (Rs)	$P_n Q_0$	$P_0 Q_0$
	Q_0	P_0	Q_1	P_1		
Wheat	100	10	120	13	1300	1000
Rice	50	100	40	110	5500	5000
Cotton	35	148	40	160	5600	5180
Butter	12	600	10	750	9000	7200
Mutton	48	500	35	600	28800	24000
Total	-	-	-	-	50200	42380

$$I_L = \frac{\Sigma P_n Q_0}{\Sigma P_0 Q_0} \times 100$$

$$I_L = \frac{50200}{42380} \times 100$$

$I_L = 118.4521$ Ans.

Example 2:

Commodity	2010		2015		$P_0 Q_0$	$P_0 Q_1$
	Quantity (kg) Q_0	Price/ kg (Rs) P_0	Quantity (kg) Q_1	Price/ kg (Rs) P_1		
Wheat	100	10	120	13	1000	1200
Rice	50	100	40	110	5000	4000
Cotton	35	148	40	160	5180	5920
Butter	12	600	10	750	7200	6000
Mutton	48	500	35	600	24000	17500
Total	-	-	-	-	42380	34620

Paasche's Formula: $$\frac{\Sigma P_0 Q_0}{\Sigma P_0 Q_1} \times 100$$

$$= \frac{42380}{34620} \times 100$$

$$= 122.41$$ Ans.

Example 3:

Commodity	2010 Quantity Q_0 (kg)	2010 Price/ kg P_0 (Rs)	2015 Quantity Q_1 (kg)	2015 Price/ kg P_1 (Rs)	P_1Q_1	P_0Q_0
Wheat	100	10	120	13	1560	1000
Rice	50	100	40	110	4400	5000
Cotton	35	148	40	160	6400	5180
Butter	12	600	10	750	7500	7200
Mutton	48	500	35	600	21000	24000
Total	-	-	-	-	40860	42380

P_1Q_0	P_0P_1
13x100 = 1300	10x120 = 1200
110x5 = 5500	100x40 = 4000
160x35 = 5600	148x40 = 5920.
750x12 = 9000	600 x10 = 6000
600x48 = 28800	500x35 = 17500
Total = 50200	Total = 34620

Paasche's $= \frac{P_1Q_1}{P_0Q_0} = \frac{40860}{42380}$

Paasche's Index $= 0.9641$

Question 7a:

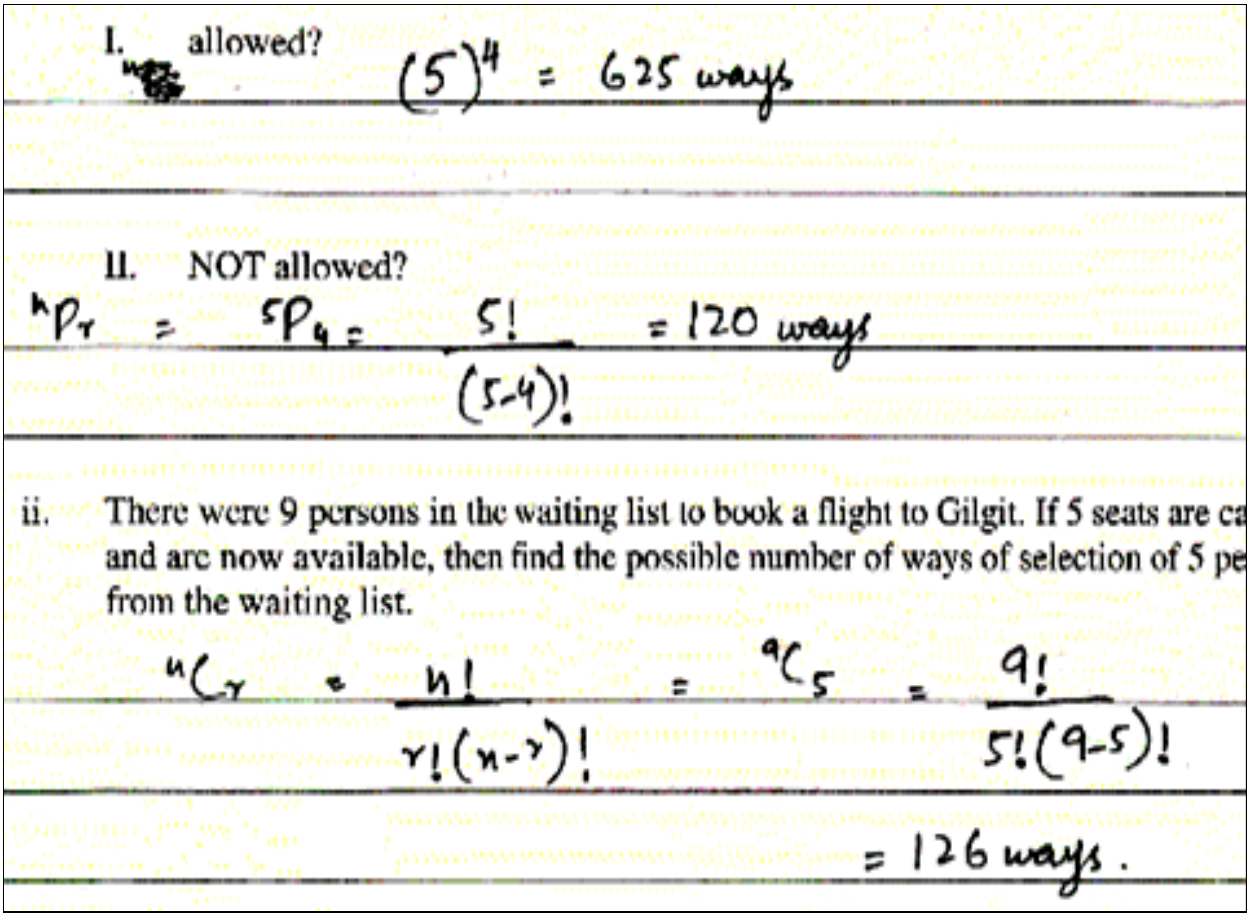
i. How many passwords of four letters are possible by using the letters of the word 'SOUTH', if repetition is

   I. allowed?
   II. NOT allowed?

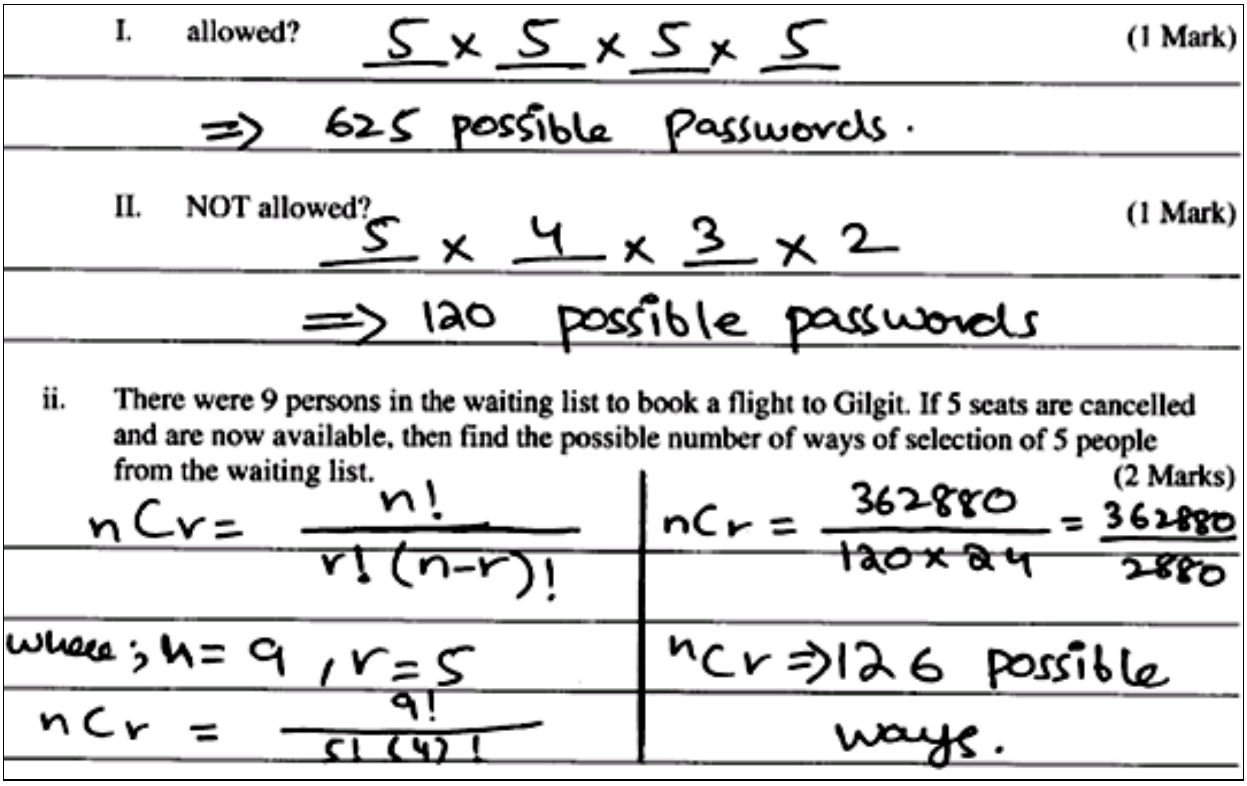
ii. There were 9 persons in the waiting list to book a flight to Gilgit. If 5 seats are cancelled and are now available, then find the possible number of ways of selection of 5 people from the waiting list.

Better responses exhibited that for part **i**, candidates found the total number of passwords of letters of the word 'SOUTH' with and without repetition. In part **ii**, candidates were able to find the possible number of ways to select 5 people out of 9 people. It showed that candidates have clear concepts of counting techniques and used appropriate formula to solve the given cases in the question.

Example 1:



Example 2:



I. allowed? $5 \times 5 \times 5 \times 5$ (1 Mark)

$\Rightarrow$ 625 possible passwords.

II. NOT allowed? $5 \times 4 \times 3 \times 2$ (1 Mark)

$\Rightarrow$ 120 possible passwords

ii. There were 9 persons in the waiting list to book a flight to Gilgit. If 5 seats are cancelled and are now available, then find the possible number of ways of selection of 5 people from the waiting list. (2 Marks)

$nCr = \frac{n!}{r!(n-r)!}$

where; $n = 9, r = 5$

$nCr = \frac{9!}{5!(4)!}$

$nCr = \frac{362880}{120 \times 24} = \frac{362880}{2880}$

$nCr \Rightarrow 126$ possible ways.

Weaker responses reflected ambiguity in the concepts of counting technique. Most of the candidates were failed to do part **i** and part **ii**. The most frequent mistakes noted is the wrong selection and application of formulae in different cases which is evident in the given example.

Example 1:

3125

II. NOT allowed?

120

ii. There were 9 persons in the waiting list to book a flight to Gilgit. If 5 seats are and are now available, then find the possible number of ways of selection of 5 from the waiting list.

total person = 9

possible seat = 5

possible number = 15120

Example 2:

I. allowed? (1 Ma

5 x 5 x 5 x 5 x 5 = 3125

II. NOT allowed? (1 Ma

5 x 4 x 3 x 2 x 1 = 120

There were 9 persons in the waiting list to book a flight to Gilgit. If 5 seats are cancelled and are now available, then find the possible number of ways of selection of 5 people from the waiting list. (2 Mar

9/5 = 1.8 possible number of ways

Question 7b:

A bag contains eleven cards. Each letter of the word MATHEMATICS is written on a separate card kept in the bag. If two cards are randomly drawn without replacement, then find the probability that

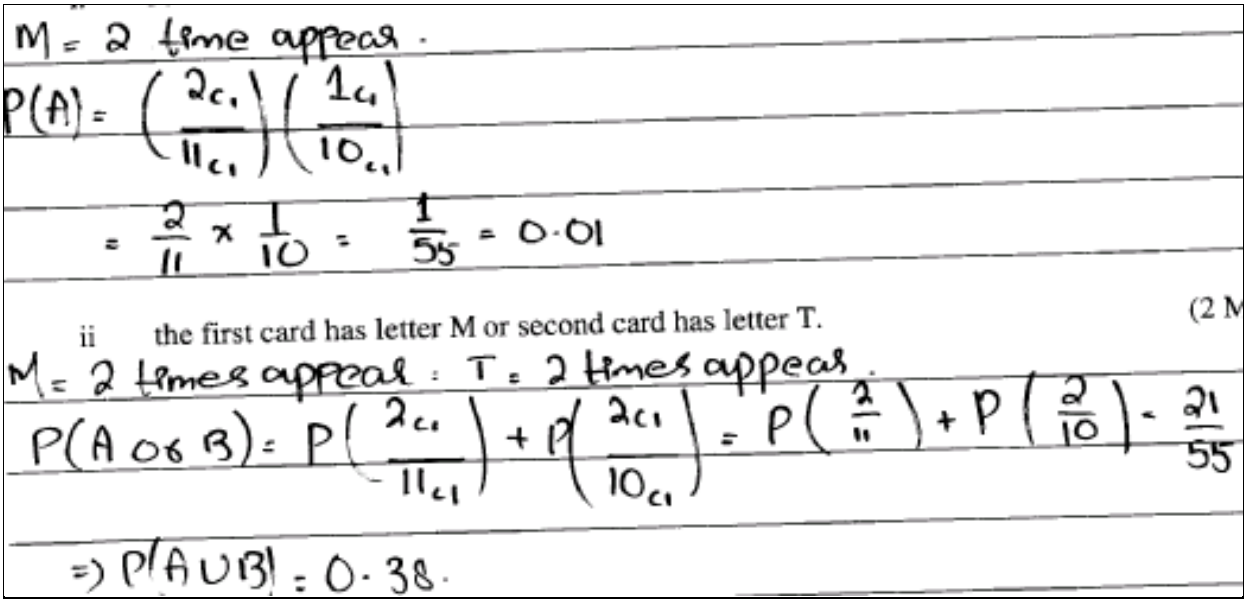
i. both cards have letter M.

ii the first card has letter M or second card has letter T.

There were choice between part **a** and part **b**. But mostly attempted part **a** of the question.

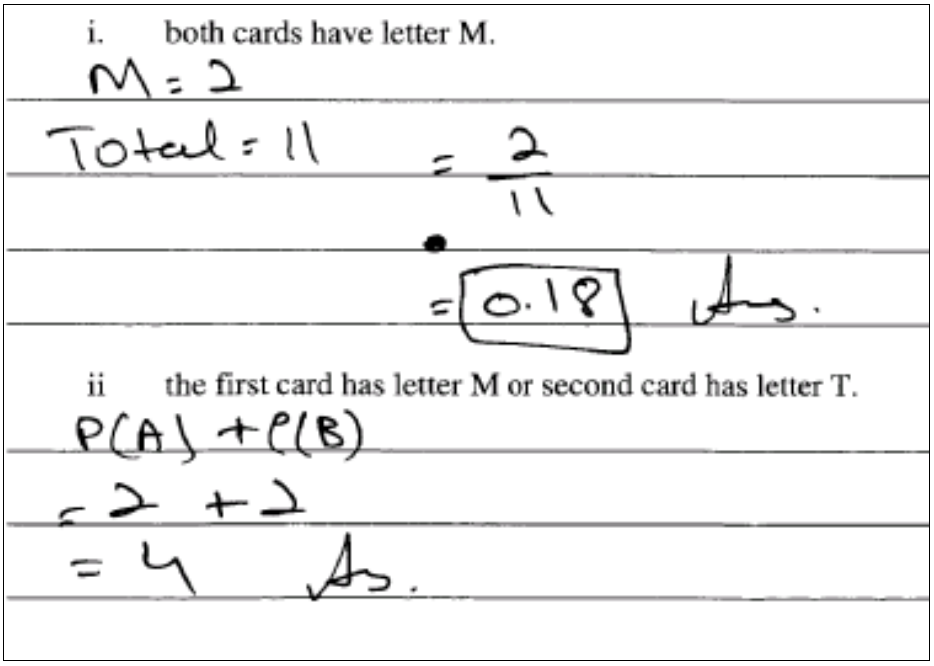
Better responses indicated that the candidates were able to understand the given situation and were able to find the probabilities correctly. Part **i** and part **ii** was based on the probability under different conditions. The candidates found the probability and translated the given conditions successfully to exhibit their command over the concepts of probabilities.

Example 1:



Weaker responses displayed that many candidates failed to understand the given situation. It is indicative of the fact that candidates have misconceptions in finding probabilities of the events. Probability of event cannot be more than 1 but weaker responses reflected that candidates did not know this basic axiom of probability. As a result, they were failed to find the required probabilities.

Example 1:



Example 2:

